

DeMoP: A Language-Model-guided Mixture-of-Experts Framework for Cancer Prognosis

Chen Tang^{1,4}, Lei Yu^{1,4}, Qiwei Li², Lin Xu^{1,3 *}

¹Quantitative Biomedical Research Center, Department of Health Data Science & Biostatistics, Peter O'Donnell Jr. School of Public Health, University of Texas Southwestern Medical Center, Dallas, TX, U.S.A

²Department of Mathematical Sciences, The University of Texas at Dallas, Richardson, Texas, U.S.A

³Department of Pediatrics, Division of Hematology/Oncology, University of Texas Southwestern Medical Center, Dallas, TX, U.S.A

⁴These authors contributed equally

*Corresponding Authors:

Lin Xu, PhD (Lin.Xu@UTSouthwestern.edu)

Abstract

Accurate prediction of clinical outcomes from clinical-omic data remains a key challenge in precision medicine. We present DeMoP, a language-model-guided mixture-of-experts framework that converts patient clinical and molecular profiles into structured natural-language representations for pan-cancer prognosis. Using a fine-tuned DeBERTa-v3-large backbone, attention-based pooling, and a ResNet-based mixture-of-experts head, DeMoP models nonlinear patient-specific patterns from clinical variables, copy-number alterations, and gene descriptions. Across two large-scale pan-cancer cohorts, DeMoP outperformed conventional machine-learning and deep-learning baselines. At the cohort level, it achieved AUROCs of 0.939 and 0.805, with corresponding Youden indices of 0.673 and 0.475. In transfer analysis, the model trained on a large clinical registry retained predictive utility in an independent smaller cohort. Gene-level ablations recovered experimentally validated cancer genes and prioritized understudied candidates. DeMoP therefore provides a scalable framework for prognostic risk stratification and biological hypothesis generation from heterogeneous clinical-omic data.

Keywords: clinical multimodal model; mixture-of-experts; language modeling; cancer prognosis;

Introduction

Accurate cancer prognosis is essential for treatment selection, clinical trial stratification, and long-term disease management. However, predicting patient outcomes remains challenging because clinical and molecular data are intrinsically heterogeneous^{1,2}. Clinical variables such as age, sex, and tumor stage are relatively low-dimensional and interpretable, whereas molecular profiles, including somatic mutations, copy-number alterations, gene expression and epigenetic features, are high-dimensional and complex^{3,4}.

Traditional statistical and machine-learning approaches often rely on domain-specific feature engineering to integrate these heterogeneous modalities. Such approaches are labor-intensive and may miss nonlinear, higher-order interactions between clinical and genomic factors that underlie patient outcomes^{5,6}. Recent advances in natural language processing (NLP) and pretrained language models, including encoder-based models such as RoBERTa⁷ and DeBERTa⁸, offer a different strategy for modeling biomedical data. By representing structured patient information as natural-language-like sequences, transformer architectures can learn cross-modal relationships within a shared representation space⁹. In this setting, heterogeneous clinical and molecular attributes are serialized into token sequences and processed by self-attention, enabling flexible integration without bespoke feature engineering.

General-purpose biomedical large language models such as BioGPT¹⁰ and Med-PaLM¹¹ have shown strong performance in medical question answering, clinical reasoning, and text understanding. However, these models are primarily designed for unstructured clinical text and are not optimized for structured multi-omics or tabular patient data, which limits their applicability to integrative prognostic modeling. In parallel, transformer-based approaches for structured and tabular data, including TabPFN¹², TabTransformer¹³ and FT-Transformer¹⁴, have improved performance over conventional machine-learning methods by modeling feature interactions with self-attention. Although effective for tabular prediction, these approaches usually assume fixed feature schemas and do not readily accommodate heterogeneous biomedical modalities or leverage large-scale pretrained language representations.

More recently, LLM-inspired approaches have been extended to biological sequences and single-cell data. Models such as scGPT¹⁵, CELLama¹⁶, and sciLaMA¹⁷ adapt transformer architectures

to encode gene expression profiles and molecular features into tokenized representations, enabling downstream tasks such as cell-type annotation and perturbation modeling. Despite their promise, these approaches primarily focus on cell-level problems and have not been systematically evaluated for patient-level outcome prediction across heterogeneous real-world patient cohorts.

To address these challenges, we present DeMoP, a language-guided framework that combines a fine-tuned DeBERTa-v3-large encoder (24 transformer layers, with 304M backbone parameters and 131M embedding parameters)^{8,18}, leveraging concatenated representations from its final four hidden layers, with an attention-based pooling module and a ResNet-based mixture-of-experts head for cancer prognosis. DeBERTa’s disentangled attention and relative positional encoding support fine-grained modeling of structured patient representations, whereas the MoE head enhances the model’s adaptability through input-dependent specialization over heterogeneous data patterns. Unlike prior approaches restricted to single cancer types or more homogeneous datasets, DeMoP is designed for cross-cohort generalization and heterogeneous clinical and omics data integration. We evaluate DeMoP on multiple independent cancer patient cohorts, including AACR Project GENIE¹⁹ and the TCGA database²⁰ and benchmark it against conventional machine-learning methods and deep-learning architectures.

Results

Overview of DeMoP

DeMoP encodes patient-level clinical characteristics (e.g., sex and age) and gene-level information (e.g., copy-number alterations and brief gene descriptions) as unified structured natural-language representations (**Fig. 1**), enabling heterogeneous clinical and molecular data to be processed within a common framework.

These representations are passed through a fine-tuned DeBERTa-v3-large encoder (24 transformer layers, with 304M backbone parameters and 131M embedding parameters). Concatenated representations from its final four hidden layers are then used to aggregate complementary contextual information across semantic depths, conceptually inspired by YOLOv2-style feature aggregation²¹. Together, these components form the clinical and molecular embedding module of DeMoP.

Patient embeddings are then passed to a mixture-of-experts (MoE) head, in which a gating network assigns input-dependent weights to multiple specialized expert subnetworks according to each sample's embedding pattern. This design enables adaptive modeling of heterogeneous patient representations and complementary nonlinear patterns that may not be fully captured by a single prediction head. Together with the contextual patient embeddings, the MoE head constitutes the prediction module of DeMoP (**Methods: Mixture-of-experts head**).

To validate the application of DeMoP in the real-world clinical-omic datasets, we conducted three complementary computational experiments: (1) training and evaluation on the AACR Project GENIE cohorts with 63,090 patients, (2) training and evaluation on the TCGA cohorts with 4,123 patients, and (3) transfer analyses in which models was trained on a larger clinical cohort (i.e., AACR project GENIE cohorts) and then evaluated on a smaller cohort with same cancer types (i.e., TCGA cohorts). Across these settings, we compared the performance of DeMoP with several conventional machine-learning and deep-learning approaches.

DeMoP achieves strong predictive performance on the GENIE cohorts

As a large real-world oncology patient cohort, AACR Project GENIE provides a rigorous testbed for evaluating performance across 84 diverse cancer types (**Method: Dataset Collection and Preprocessing**). For the GENIE cohort, samples from patients who were alive but had been followed for less than three years were excluded. This filtering reduced the number of samples from 130,124 to 64,681. Samples with zero expression across all genes in the 3-year mortality group were then removed. After quality-control filtering, 63,090 samples were included in the final GENIE cohort. Using the GENIE cohort ($N = 63,090$), we trained DeMoP on 80% training set and evaluated it on the remaining 20% held-out internal test set. **Fig. 2A** compares the performance of DeMoP with six state-of-the-art methods on the GENIE test set after applying predefined survival-based inclusion criteria. Focused on 3-year survival prediction, DeMoP achieved an F1 score of 0.72, outperforming default parameter-setting baseline models: logistic regression (0.51), random forest (0.58), XGBoost (0.66), multilayer perception (0.57), CNN (0.63), and LSTM (0.59) with the identical training and testing datasets as DeMoP. The consistent margin over both shallow classifiers and other deep architectures suggests that DeMoP captures complementary information from high-dimensional gene copy-number alteration profiles and diverse clinical annotations. For large-scale patient cohorts aggregated across multiple hospitals or medical centers, DeMoP's integrative capacity is important for modeling the heterogeneity inherent in real-world clinical data.

Fig. 2B shows F1 scores across the top eight sample-size cancer types within the GENIE cohort. In common solid tumors such as breast cancer ($n = 1,682$), non-small cell lung cancer ($n = 1,596$) and colorectal cancer ($n = 1,064$), DeMoP improved F1 score over random forest (0.65 versus 0.57 in breast cancer, 0.71 versus 0.56 in non-small cell lung cancer and 0.63 versus 0.59 in colorectal cancer). In glioma ($n = 658$), DeMoP achieved an F1 of 0.89, compared with 0.79 for random forest. In prostate cancer ($n = 567$), DeMoP achieved an F1 score of 0.70, compared with 0.63 for random forest and 0.61 for logistic regression. In mature B-cell neoplasms ($n = 533$), DeMoP more than doubled logistic regression's F1 score (0.44 versus 0.08). Smaller gains in ovarian cancer (0.50 versus 0.47) and bladder cancer (0.62 versus 0.50) further support robust performance across solid and hematological malignancies.

To understand possible biological relevance, we performed a systematic gene-level ablation analysis using DeMoP (**Methods: Ablation Study**). For each gene, we removed the corresponding

gene-level segment, including its copy-number alteration data and associated brief descriptive text, from patient descriptions, passed the modified inputs through the fine-tuned model, and measured the resulting change in predicted probability for the model's original predicted class relative to the full input. We defined each gene's importance score as the mean reduction in prediction confidence upon omission of its gene-level information across all patients in the cohort (**Methods: Ablation Study, Eq. 21**); a larger importance score indicates stronger model dependence on that gene's encoded copy-number and descriptive information. The 20 genes with the highest mean importance scores are shown in **Fig. 2C**. MYC had the highest importance score (0.12), consistent with its central role in cell proliferation across cancer types²². CDKN2B (0.07) also ranked highly, in line with its role in cell-cycle control^{23,24}. ALOX12B (0.07), a lipoxygenase implicated in ferroptosis and inflammation²⁵, emerged unexpectedly as a top feature, suggesting that the model may capture underexplored microenvironmental dependencies. SYK (0.05), a kinase in B-cell receptor signaling²⁶, ranked fourth and is consistent with the strong performance of DeMoP in mature B-cell neoplasms. Other contributors included BCL6 (0.03), a key regulator of germinal-center formation and B-cell differentiation²⁷, and SMAD4 (0.02), a critical mediator of TGF- β signaling with established tumor suppressor functions²⁸.

At the cohort level, DeMoP achieved an overall AUROC of 0.939, with sensitivity 0.772, specificity 0.901, a Youden index of 0.673, accuracy 0.874, and an F1 score of 0.718. We next examined cancer-specific discrimination in tumor types with more than 200 samples in the test set (**Fig. 2D**). Among 15 cancer types, the highest performance was observed in leukemia ($n = 266$, AUROC = 0.997), followed by mature B-cell neoplasms ($n = 533$, AUROC = 0.980), pancreatic cancer ($n = 397$, AUROC = 0.961), prostate cancer ($n = 567$, AUROC = 0.960) and glioma ($n = 658$, AUROC = 0.949). Strong discrimination was also observed in endometrial cancer ($n = 422$, AUROC = 0.943), bladder cancer ($n = 440$, AUROC = 0.930), soft tissue sarcoma ($n = 417$, AUROC = 0.926), renal cell carcinoma ($n = 228$, AUROC = 0.915), breast cancer ($n = 1682$, AUROC = 0.915), non-small cell lung cancer ($n = 1596$, AUROC = 0.896), esophagogastric cancer ($n = 351$, AUROC = 0.889), colorectal cancer ($n = 1064$, AUROC = 0.882), melanoma ($n = 409$, AUROC = 0.850) and ovarian cancer ($n = 469$, AUROC = 0.845). These results indicate that DeMoP performs well at the cohort level while maintaining broad discriminatory ability across a broad spectrum of solid and hematological malignancies.

Together, these analyses show that DeMoP outperforms conventional and deep-learning methods on the GENIE cohort and shows evidence of applicability across the tumor lineages examined. The gene-level importance scores recover known pathways while also highlighting less well-studied regulators such as ALOX12B, generating testable biological hypotheses for experimental biologists and clinical investigators to follow up.

DeMoP achieves strong predictive performance on the TCGA cohort

For the TCGA cohort, the initial dataset contained 9,370 samples. Patients were categorized according to their overall survival status and survival duration. Patients who died within 36 months were assigned to the 3-year mortality group, whereas patients who remained alive for more than 36 months were assigned to the long-term survival group. Patients who did not meet either criterion were excluded from further analysis. After survival-based filtering, 4,470 samples were retained. Samples with no copy number variation (zero value) across all genes in the 3-year mortality group were subsequently removed, leaving 4,371 samples. We further excluded patients in the 3-year mortality group for whom both *PERSON NEOPLASM CANCER STATUS* and *NEW TUMOR EVENT AFTER INITIAL TREATMENT* columns were missing. The final TCGA cohort consisted of 4,123 samples. Using the TCGA cohort (N = 4,123), we trained DeMoP on 80% training set and evaluated it on the remaining 20% held-out internal test set. DeMoP achieved the highest F1 score (0.72) among all benchmarked methods, outperforming logistic regression (0.59), random forest (0.65), XGBoost (0.68), CNN (0.65), LSTM (0.61), and MLP (0.58) (**Fig. 3A**). The pan-cancer setting poses a substantial modeling challenge because tumor types differ markedly in their clinical and molecular profiles. DeMoP addresses this heterogeneity through a MoE architecture that dynamically allocates specialized experts to capture subtype-specific patterns.

We next examined performance by individual cancer type to assess model robustness. **Fig. 3B** shows results across five cancer types with at least 40 patients and at a minimum of 20 death events. In non-small cell lung cancer (N = 86), DeMoP achieved an F1 of 0.73, compared with 0.70 for random forest and 0.53 for logistic regression. In renal clear cell carcinoma (N = 64), DeMoP achieved an F1 of 0.53, more than doubling logistic regression (0.25) and exceeding random forest (0.44). Consistent improvements were also observed in head and neck cancer (DeMoP, 0.81 versus random forest, 0.73, and logistic regression, 0.68), esophagogastric cancer (0.89 versus 0.77 and 0.78) and bladder cancer (0.79 versus 0.73 and 0.76). These results indicate that DeMoP maintains

predictive accuracy even in smaller and more heterogeneous subcohorts, underscoring its potential utility for cancer types where conventional approaches struggle.

The importance score of genes was calculated, and the top 20 genes were extracted for the study (**Fig. 3C**). CDKN2B again ranked highest (importance score = 0.06), reinforcing its consistent contribution across datasets. SMAD4 (0.02) remained among the top features. Canonical oncogenes, including EGFR²⁹ and MYC²² (each 0.01), were also retained, indicating that DeMoP captures shared driver signals despite cohort heterogeneity. Additional genes, such as ASXL1³⁰, BCL6²⁷, and APC³¹ (≤ 0.01), further highlight the model's ability to integrate diverse oncogenic programs.

At the cohort level, DeMoP achieved an overall AUROC of 0.805, with sensitivity 0.791, specificity 0.684, a Youden index of 0.475, accuracy 0.729 and an F1 score of 0.712. We further examined cancer-specific discrimination in the test set and identified three cancer types with more than 20 samples and AUROC values above 0.8. Among these, thyroid cancer showed the highest performance ($n = 39$, AUROC = 0.947), followed by glioma ($n = 36$, AUROC = 0.835) and breast cancer ($n = 82$, AUROC = 0.831) (**Fig. 3D**). These results indicate that DeMoP performs robustly at the cohort level and retains discriminative ability across multiple cancer types.

Collectively, these analyses show that DeMoP surpasses both conventional and deep-learning baselines in the TCGA pan-cancer cohorts while generalizing across individual tumor subtypes. The gene-level importance scores align with established cancer biology, supporting the model's interpretability and translational relevance.

DeMoP transfers predictive utility from large to small cohorts

Cross-dataset validation is essential for testing whether a model trained in one cohort generalizes to an independent dataset with different sequencing platforms, annotation pipelines, and patient populations. We therefore trained DeMoP on 80% of AACR Project GENIE ($N = 50,417$) and evaluated it directly on the full filtered TCGA cohort ($N = 4,123$ samples). The overall transfer F1 score was calculated using all 4,123 TCGA samples, and Figure 4A also shows cancer-specific results for the eight cancer types with the largest sample sizes. Because GENIE is an order of magnitude larger than TCGA, we conducted this one-directional GENIE-to-TCGA transfer to

examine whether representations learned from a large clinical registry (e.g., GENIE) retain predictive utility in an independent cohort (e.g., TCGA).

Transfer performance varied by tumor lineage: bladder cancer (C5, $n = 220$) transferred best ($F1 = 0.77$), followed by hepatobiliary cancer (C7, $n = 157$; $F1 = 0.73$) and esophagogastric cancer (C4, $n = 245$; $F1 = 0.72$). Non-small cell lung cancer (C1, $n = 424$) and head and neck cancer (C3, $n = 272$) also transferred reasonably well ($F1 = 0.66$ and 0.71 , respectively). Ovarian epithelial tumor (C8, $n = 153$) showed an $F1$ score of 0.57 . By contrast, renal clear cell carcinoma (C2, $n = 315$) and melanoma (C6, $n = 197$) showed the largest drops in performance ($F1 = 0.42$ and 0.49 , respectively). Across all cancer types, the transferred model achieved an overall $F1$ score of 0.62 (**Fig. 4A**).

Comparing these cross-cohort results with within-cohort TCGA performance (**Fig. 3A, B**) reveals two consistent patterns: First, the overall $F1$ of 0.62 in the GENIE-to-TCGA transfer setting is lower than the 0.72 observed in the TCGA self-test setting, suggesting the impact of cohort-specific differences and batch effects. Second, tumor types that performed well in TCGA self-testing, such as bladder cancer (0.79 with the TCGA-trained model; 0.77 with the GENIE-trained model) and esophagogastric cancer (0.89 with the TCGA-trained model; 0.72 with the GENIE-trained model), also transferred relatively well. Conversely, tumor types with greater molecular heterogeneity or annotation discordance, such as renal carcinoma and melanoma, showed the largest performance degradation. These patterns suggest that, in this GENIE-to-TCGA setting, DeMoP learns transferable biological representations and may benefit from domain adaptation when applied to new clinical registries.

When examining biological relevance in DeMoP, CDKN2B again ranked first (impact score = 0.05), reinforcing its consistent importance across datasets. MYC (0.04) remained among the top features, indicating stable recovery of core oncogenic candidate genes. In contrast, FOXL2³² (0.01) and SYK (0.01) were more prominent in the transfer setting, suggesting a potential role in enhancing cross-cohort generalization. Additional contributors, including BCL6, GNAQ, and ALOX12B, further support the model's ability to capture both conserved and context-dependent signals.

Comparing importance scores across the GENIE-only (**Fig. 2C**), TCGA-only (**Fig. 3C**), and GENIE-to-TCGA ablation analyses (**Fig. 4B**) reveals both stable and context-dependent gene

signals. CDKN2B and MYC consistently ranked among the top candidate genes across all settings, indicating robust recovery of core oncogenic programs. In contrast, SMAD4 showed higher importance in cohort-specific analyses (importance score = 0.02 in both GENIE and TCGA) than in the transfer setting (0.006), suggesting reduced contribution under cross-cohort generalization. Notably, FOXL2 and SYK gained importance during transfer, suggesting they may capture features that enhance cross-cohort robustness. Conversely, cohort-enriched cancer genes such as PIK3CA³³ and MAP3K1³⁴ ranked lower in the transfer analysis, consistent with differences in gene copy-number alteration prevalence and cohort-specific genomic composition.

In the clinical settings, training in a larger registry conferred strong predictive power, but direct application to a smaller independent cohort remained challenging. Our results show that DeMoP is promising to solve this clinical challenge by learning transferable representations of core cancer genes and biological processes. It is worth noting that optimal cross-cohort performance may require domain adaptation to reconcile platform- and annotation-related differences. In summary, these insights lay the groundwork for robust deployment of DeMoP in heterogeneous precision-oncology settings.

Discussion

DeMoP introduces a language-guided mixture-of-experts framework for disease prognosis that unifies structured clinical variables and molecular data through natural-language representations, and captures nonlinear interactions across heterogeneous modalities without manual feature engineering.

Across two large-scale and independent cohorts, AACR Project GENIE and TCGA, DeMoP consistently outperformed conventional machine-learning methods and deep-learning architectures, achieving robust predictive performance across diverse cancer types. Importantly, DeMoP is capable of transferring predictive utility from large to small cohorts, highlighting its potential for deployment in real-world, heterogeneous clinical settings. Beyond predictive performance, DeMoP provides biologically meaningful interpretability through gene-level attribution analyses. The identification of established oncogenic candidate genes, such as CDKN2B and MYC, and less characterized candidates, such as ALOX12B, suggests that the model captures both canonical and potentially novel mechanisms underlying tumor progression. This dual capacity for prediction and hypothesis generation positions DeMoP as a valuable tool not only for clinical risk stratification but also for biological discovery.

Despite these strengths, several limitations warrant consideration. DeMoP currently integrates clinical and molecular features but does not yet incorporate additional modalities such as transcriptomics, proteomics or longitudinal clinical trajectories. Although cross-cohort generalization was encouraging, performance varied across tumor types, indicating that domain adaptation or calibration may be needed to address dataset-specific biases. In addition, serializing structured patient features as natural language may obscure some fine-grained quantitative relationships, motivating future hybrid models that combine language-based and structured representations.

Several directions could further extend the DeMoP framework. Incorporating additional multimodal data sources, including imaging and single-cell omics, may yield more comprehensive patient representations. Domain-adaptation or continual-learning strategies may improve robustness across institutions and populations. Larger and more diverse training datasets may also

help the model capture rare but clinically important patterns. Finally, prospective clinical validation will be necessary to establish the utility of DeMoP as a decision-support tool.

Overall, DeMoP shows that language-model-based architectures can provide a unified framework for integrating clinical and molecular data for cancer prognosis. As foundation models continue to evolve, approaches such as DeMoP may help bridge complex biomedical data and actionable clinical insights.

Methods

Dataset Collection and Preprocessing

AACR Project GENIE data¹⁹ were downloaded from Synapse (<https://www.synapse.org/Synapse:syn7222066/wiki/405659>). TCGA²⁰ data were obtained from cBioPortal (<https://www.cbioportal.org/>). Copy-number-variation data from both TCGA and AACR Project GENIE were available as gene-level annotations and required no additional reformatting. For AACR Project GENIE, overall survival time was defined as the interval between the sample sequencing date and the censoring date. For TCGA, overall survival was obtained directly from clinical annotations. Patients were then assigned binary prognostic labels based on 3-year overall survival status. Patients who died within 3 years were labeled as class 1, whereas patients who survived for more than 3 years were labeled as class 0.

Text representation

For each patient, we define the prompt string $S^{(i)}$ as the concatenation of the basic clinical information and the gene profile. For example,

$$\begin{aligned} S^{(i)} = & \text{CLINICAL INFORMATION: } h_1^{(i)}, h_2^{(i)}, \dots, h_n^{(i)}; \\ & \text{GENE PROFILE: } g_1^{(i)}, g_2^{(i)}, \dots, g_m^{(i)}. \end{aligned} \quad (1)$$

Here, $h_1^{(i)}, h_2^{(i)}, \dots, h_n^{(i)}$ denotes the basic clinical information and $g_1^{(i)}, g_2^{(i)}, \dots, g_m^{(i)}$ denotes the set of observed gene-level segments, including copy-number alteration data and associated brief descriptive text, for a given patient. Formatting the data in this way places numerical and categorical information, together with brief gene descriptions, in a single token sequence and allows the language model to learn cross-modal dependencies directly from text.

Patient Data Encoding

The string $S^{(i)}$ in equation (1) is tokenized by the DeBERTa-V3 tokenizer¹⁸ into a sequence of token indices

$$t^{(i)} = (t_1^{(i)}, t_2^{(i)}, \dots, t_T^{(i)}), \quad t_s^{(i)} \in \{1, \dots, V\}, \quad s = 1, \dots, T. \quad (2)$$

along with an attention mask

$$m^{(i)} = (m_1^{(i)}, m_2^{(i)}, \dots, m_T^{(i)}), \quad m_s^{(i)} \in \{0, 1\}, \quad s = 1, \dots, T. \quad (3)$$

Here V is the tokenizer vocabulary size, $m_{t^i}=1$ indicates a real token, and $m_{t^i}=0$ indicates padding. This procedure converts each patient's mixed-modality record into a uniform sequence of discrete symbols. Each token index t^i is mapped to a d -dimensional embedding vector e_{t^i} , yielding the input matrix

$$Z^{(i)} = \begin{bmatrix} e_{t_1^{(i)}} \\ e_{t_2^{(i)}} \\ \vdots \\ e_{t_T^{(i)}} \end{bmatrix} \in \mathbb{R}^{T \times d}. \quad (4)$$

The embedding matrix $Z^{(i)}$ serves as input to the DeBERTa-V3 Transformer encoder, enabling all patient features to be processed through self-attention and feed-forward layers without separate feature-engineering steps.

DeBERTa-V3 Large encoding and layer fusion

We use the pretrained DeBERTa-V3 Large encoder as the backbone. Let Z in $\mathbb{R}^{(T \times d)}$ be the input embedding matrix for a single patient, where T is the token-sequence length and $d=1024$ is the hidden dimension of each DeBERTa layer. The Transformer produces a set of hidden-state tensors

$$\{h_t^{(l)} \in \mathbb{R}^d \mid l = 1, 2, \dots, 24; t = 1, 2, \dots, T\}. \quad (5)$$

Following the implemented architecture, we fuse the last four hidden layers at each token position by concatenation to combine information from the upper contextual layers:

$$H_t = [h_t^{(21)}, h_t^{(22)}, h_t^{(23)}, h_t^{(24)}] \in \mathbb{R}^{4d}. \quad (6)$$

This wide representation H_t combines multi-level features but has dimension $4d$, which is computationally burdensome for downstream modules. To reduce dimensionality while encouraging cross-layer interactions, we apply a single-layer MLP with GELU activation³⁵ to each token representation:

$$z_t = \text{GELU}(W_p H_t + b_p), \quad W_p \in \mathbb{R}^{d \times 4d}, \quad b_p \in \mathbb{R}^d. \quad (7)$$

Here the GELU nonlinearity serves as a learned bottleneck and mixes the four sub-layer channels before pooling. The resulting vector z_t in $\mathbb{R}^d$ is a compact fused embedding for token t . This

projected sequence $\{z_t\}_{t=1}^T$ is then passed to the attention-based pooling and Mixture-of-Experts head.

Attention-Based Token Pooling

After projecting each token embedding to z_t in $\mathbb{R}^d$, we aggregate the sequence $\{z_t\}_{t=1}^T$ into a single vector v in $\mathbb{R}^d$ via a lightweight self-attention mechanism⁹. We first compute query, key and value matrices by linear projections of the token embeddings:

$$Q = ZW_q, \quad K = ZW_k, \quad V = ZW_v. \quad (8)$$

where Z in $\mathbb{R}^{(T \times d)}$ stacks the z_t vectors, and W_q, W_k, W_v in $\mathbb{R}^{(d \times d)}$ are learned projection matrices.

We then form the $T \times T$ attention weight matrix by scaled dot-product attention followed by SoftMax normalization:

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \quad A \in \mathbb{R}^{T \times T}. \quad (9)$$

We next compute the context matrix and apply average pooling over tokens:

$$C = AV, \quad v = \frac{1}{T} \sum_{t=1}^T C_t. \quad (10)$$

where C_t denotes the t th row of C . The resulting vector v is a global patient representation and is passed to the mixture-of-experts head for final classification.

Mixture-of-Experts (MoE) Head

To capture heterogeneous prognostic patterns across patients, we route the pooled representation v in $\mathbb{R}^d$ through K parallel expert networks via a soft-gating mechanism. In our implementation, $K=4$. The gating module is a two-layer MLP with layer normalization, GELU activation and dropout:

$$\begin{aligned} \tilde{v}_g &= \text{Dropout}\left(\text{GELU}\left(\text{LN}(W_{g1}v + b_{g1})\right)\right). \\ u &= W_{g2}\tilde{v}_g + b_{g2}, \quad u \in \mathbb{R}^K. \end{aligned} \quad (11)$$

The gating scores are normalized into a distribution over experts:

$$g_k = \frac{\exp(u_k)}{\sum_{j=1}^K \exp(u_j)}, \quad k = 1, \dots, K. \quad (12)$$

Each expert k applies two sequential ResNet-style fully connected residual blocks to the shared input v . The first block maps the representation from d to h dimensions:

$$\begin{aligned} a_k^{(1)} &= \text{Dropout}\left(\text{GELU}\left(W_{k,1}^{(1)}v + b_{k,1}^{(1)}\right)\right). \\ \tilde{r}_k^{(1)} &= \text{Dropout}\left(W_{k,2}^{(1)}a_k^{(1)} + b_{k,2}^{(1)}\right). \\ r_k^{(1)} &= \text{GELU}\left(\tilde{r}_k^{(1)} + W_{k,s}^{(1)}v + b_{k,s}^{(1)}\right). \end{aligned} \quad (13)$$

The second residual block maps the representation from h to m dimensions:

$$\begin{aligned} a_k^{(2)} &= \text{Dropout}\left(\text{GELU}\left(W_{k,1}^{(2)}r_k^{(1)} + b_{k,1}^{(2)}\right)\right). \\ \tilde{r}_k^{(2)} &= \text{Dropout}\left(W_{k,2}^{(2)}a_k^{(2)} + b_{k,2}^{(2)}\right). \\ r_k^{(2)} &= \text{GELU}\left(\tilde{r}_k^{(2)} + W_{k,s}^{(2)}r_k^{(1)} + b_{k,s}^{(2)}\right). \end{aligned} \quad (14)$$

Here h and m denote the intermediate hidden dimensions, with $h=512$ and $m=256$ in the implemented model. The shortcut projections are used because the feature dimensions change across residual blocks. Each expert then produces class logits

$$\ell_k = W_{k,\text{out}}r_k^{(2)} + b_{k,\text{out}}, \quad \ell_k \in \mathbb{R}^C. \quad (15)$$

where $C=2$ is the number of survival-group classes. Finally, the gated ensemble combines expert outputs into the overall logits

$$\ell = \sum_{k=1}^K g_k \ell_k. \quad (16)$$

which are passed through SoftMax to obtain class probabilities. The residual shortcuts stabilize gradient flow and encourage each block to learn incremental refinements for survival-group prediction.

Training Objective and Optimization

We train the entire network end-to-end by minimizing a supervised classification loss. Let y_i denote the ground-truth class label for patient i and let $\ell_{i,c}$ denote the model logit for class c . The standard cross-entropy loss over N training samples is

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\ell_{i,y_i})}{\sum_{c=1}^C \exp(\ell_{i,c})}. \quad (17)$$

Model parameters are optimized with AdamW algorithm³⁶, which combines adaptive moment estimation with decoupled weight decay. We use a linear learning-rate schedule with warmup: the learning rate increases linearly from zero to α_{max} over the first W steps and then decays linearly to zero over the remaining training steps. The best checkpoint is selected based on validation weighted F1. In the implemented training callback, early stopping is monitored using validation loss.

During evaluation, we report accuracy, weighted F1 score and AUROC. For class c ,

$$F_{1,c} = 2 \frac{P_c R_c}{P_c + R_c}. \quad (18)$$

where P_c and R_c are the precision and recall for class c . The weighted F1 score is computed as

$$F_{1,weighted} = \sum_{c=1}^C \frac{n_c}{N} F_{1,c}. \quad (19)$$

where n_c is the number of samples belonging to class c . This metric accounts for class imbalance by weighting each class-specific F1 score according to class support.

Ablation Study

We also performed a systematic gene-level ablation analysis to quantify how strongly DeMoP predictions depend on each gene-level segment under the specified input-perturbation scheme. For a given patient i and gene g_i , we removed the corresponding gene-level segment, including copy-number alteration information and associated brief descriptive text, when that segment was present. Let $\hat{y}_i$ denote the class predicted by the model from the original full input (i.e., the class with the highest predicted probability).

The modified input was then passed through the fine-tuned model, and we measured the signed change in predicted probability for this same original predicted class relative to the full input:

$$\Delta p_j^{(i)} = p_{\theta}(\hat{y}_i | S^{(i)}) - p_{\theta}(\hat{y}_i | S_{-g_j}^{(i)}). \quad (20)$$

where $p_{\theta}(\hat{y}_i | S^{(i)})$ is the probability assigned to the model's original predicted class from the full patient description, and $p_{\theta}(\hat{y}_i | S_{-g_j}^{(i)})$ is the probability assigned to that same class after removing gene-level segment g_j . The original predicted class is fixed before ablation, even if the predicted class changes after gene removal. Positive values of $\Delta p_j^{(i)}$ indicate that ablation decreases confidence in the original prediction, whereas negative values indicate increased confidence after ablation. We compute the aggregate gene-level importance score as the mean signed change across patients containing that gene-level segment:

$$I_j = \frac{1}{N_j} \sum_{i: g_j \in S^{(i)}} \Delta p_j^{(i)}. \quad (21)$$

where N_j is the number of patients whose descriptions contain the gene-level segment g_j . Genes were ranked by I_j in descending order. Larger positive scores therefore identify gene-level features whose removal, on average, most decreases the model's confidence in its original prediction. This ranking quantifies model dependence under the specified input-perturbation scheme and does not establish biological mechanism or general molecular-driver status.

Data availability

The datasets used in this study are publicly available, and the original publications for each dataset are cited in the Methods section. The source code for DeMoP can be downloaded from GitHub (<https://github.com/Lin-Xu-lab/DeMoP>)

Code Availability

The DeMoP package is available at <https://github.com/Lin-Xu-lab/DeMoP>.

Acknowledgments

The resources of the high-performance computing environment from the Quantitative Biomedical Research Center (QBRC) and BioHPC at UT Southwestern Medical Center are gratefully acknowledged.

Funding

This work was supported by the following funding: the Rally Foundation, Children's Cancer Fund (Dallas), the HHOW Award, the Sam Day Foundation Award, the Cancer Prevention and Research

Institute of Texas (RP180319, RP200103, RP220032, RP170152 and RP180805), and the National Institutes of Health funds (R01DK127037, R01CA263079, R21CA259771, R21CA273282, P30CA142543, UM1HG011996, R01NS142141, R01CA284591, R01DK130961, and R01HL144969) (to L.X.).

Contributions

CT, LY, and LX conceived and designed the study. CT and LY generated the scripts and GitHub page, and performed the data analysis. LX acquired the funding. CT, LY, QL, and LX wrote and revised the manuscript. All authors have read, revised, and approved the final manuscript.

Corresponding authors

Correspondence to Lin Xu.

Ethics declarations

All the authors declare that they have no competing interests.

References

- 1 Marusyk, A. & Polyak, K. Tumor heterogeneity: causes and consequences. *Biochimica et Biophysica Acta (BBA)-Reviews on Cancer* **1805**, 105-117 (2010).
- 2 Dagogo-Jack, I. & Shaw, A. T. Tumour heterogeneity and resistance to cancer therapies. *Nature reviews Clinical oncology* **15**, 81-94 (2018).
- 3 Hoadley, K. A. *et al.* Multiplatform analysis of 12 cancer types reveals molecular classification within and across tissues of origin. *Cell* **158**, 929-944 (2014).
- 4 Hasin, Y., Seldin, M. & Lusis, A. Multi-omics approaches to disease. *Genome biology* **18**, 83 (2017).
- 5 Kourou, K., Exarchos, T. P., Exarchos, K. P., Karamouzis, M. V. & Fotiadis, D. I. Machine learning applications in cancer prognosis and prediction. *Computational and structural biotechnology journal* **13**, 8-17 (2015).
- 6 Ching, T. *et al.* Opportunities and obstacles for deep learning in biology and medicine. *Journal of the royal society interface* **15**, 20170387 (2018).
- 7 Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint arXiv:1907.11692*.
- 8 He, P., Liu, X., Gao, J. & Chen, W. Deberta: Decoding-enhanced bert with disentangled attention. *arXiv preprint arXiv:2006.03654* (2020).
- 9 Vaswani, A. *et al.* Attention is all you need. *Advances in neural information processing systems* **30** (2017).
- 10 Luo, R. *et al.* BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Briefings in bioinformatics* **23**, bbac409 (2022).
- 11 Singhal, K. *et al.* Large language models encode clinical knowledge. *Nature* **620**, 172-180 (2023).
- 12 Hollmann, N. *et al.* Accurate predictions on small data with a tabular foundation model. *Nature* **637**, 319-326 (2025).
- 13 Huang, X., Khetan, A., Cvitkovic, M. & Karnin, Z. Tabtransformer: Tabular data modeling using contextual embeddings. *arXiv preprint arXiv:2012.06678* (2020).
- 14 Gorishniy, Y., Rubachev, I., Khrulkov, V. & Babenko, A. Revisiting deep learning models for tabular data. *Advances in neural information processing systems* **34**, 18932-18943 (2021).
- 15 Cui, H. *et al.* scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature methods* **21**, 1470-1480 (2024).
- 16 Park, J. *et al.* CELLama: foundation model for single cell and spatial transcriptomics by cell embedding leveraging language model abilities. *Advanced Science* **13**, e13210 (2026).
- 17 Hu, H., Zhang, S., Choi, Y., Malladi, V. S. & Quon, G. sciLaMA: A Single-Cell Representation Learning Framework to Leverage Prior Knowledge from Large Language Models. *bioRxiv* (2025).
- 18 He, P., Gao, J. & Chen, W. Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543* (2021).

- 19 Consortium, A. P. G. *et al.* AACR Project GENIE: powering precision medicine through an international consortium. *Cancer discovery* **7**, 818-831 (2017).
- 20 Cancer Genome Atlas Research Network, J. The cancer genome atlas pan-cancer analysis project. *Nat. Genet* **45**, 1113-1120 (2013).
- 21 Redmon, J. & Farhadi, A. in *Proceedings of the IEEE conference on computer vision and pattern recognition*. 7263-7271.
- 22 Dhanasekaran, R. *et al.* The MYC oncogene—the grand orchestrator of cancer growth and immune evasion. *Nature reviews Clinical oncology* **19**, 23-36 (2022).
- 23 Sherr, C. J. The INK4a/ARF network in tumour suppression. *Nature reviews Molecular cell biology* **2**, 731-737 (2001).
- 24 Kim, W. Y. & Sharpless, N. E. The regulation of INK4/ARF in cancer and aging. *Cell* **127**, 265-275 (2006).
- 25 Stockwell, B. R. *et al.* Ferroptosis: a regulated cell death nexus linking metabolism, redox biology, and disease. *cell* **171**, 273-285 (2017).
- 26 Mócsai, A., Ruland, J. & Tybulewicz, V. L. The SYK tyrosine kinase: a crucial player in diverse biological functions. *Nature Reviews Immunology* **10**, 387-402 (2010).
- 27 Basso, K. & Dalla-Favera, R. Roles of BCL6 in normal and transformed germinal center B cells. *Immunological reviews* **247**, 172-183 (2012).
- 28 Massagué, J. TGF β signalling in context. *Nature reviews Molecular cell biology* **13**, 616-630 (2012).
- 29 Uribe, M. L., Marrocco, I. & Yarden, Y. EGFR in cancer: signaling mechanisms, drugs, and acquired resistance. *Cancers* **13**, 2748 (2021).
- 30 Katoh, M. Functional and cancer genomics of ASXL family members. *British journal of cancer* **109**, 299-306 (2013).
- 31 Kinzler, K. W. & Vogelstein, B. Lessons from hereditary colorectal cancer. *Cell* **87**, 159-170 (1996).
- 32 Shah, S. P. *et al.* Mutation of FOXL2 in granulosa-cell tumors of the ovary. *New England Journal of Medicine* **360**, 2719-2729 (2009).
- 33 Karakas, B., Bachman, K. E. & Park, B. H. Mutation of the PIK3CA oncogene in human cancers. *British journal of cancer* **94**, 455-459 (2006).
- 34 Xue, Z. *et al.* MAP3K1 and MAP2K4 mutations are associated with sensitivity to MEK inhibitors in multiple cancer models. *Cell research* **28**, 719-729 (2018).
- 35 Hendrycks, D. & Gimpel, K. Gaussian error linear units (gelus). *arXiv preprint arXiv:1606.08415* (2016).
- 36 Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).

Figures

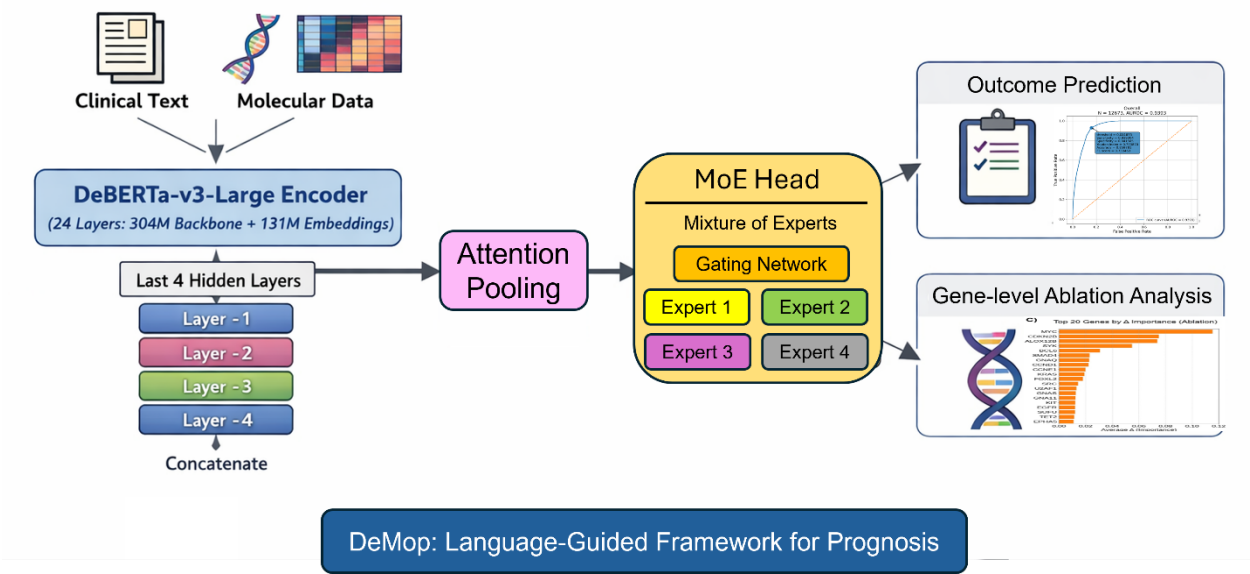


Figure 1. Schematic overview of the DeMoP framework. Patient-level clinical and molecular features are serialized into structured natural-language representations and encoded by DeBERTa-v3-large. Concatenated representations from the final four hidden layers are then processed by an attention-based pooling module and a ResNet-based mixture-of-experts prediction head for pan-cancer outcome prediction and gene-importance analysis. The number of experts shown here is illustrative only and can be adjusted.

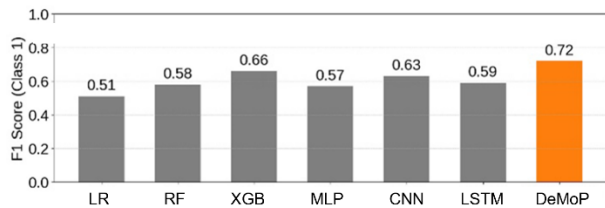
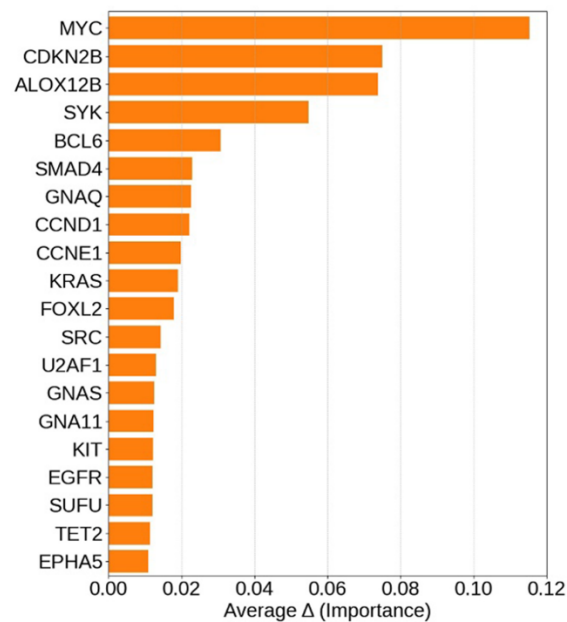
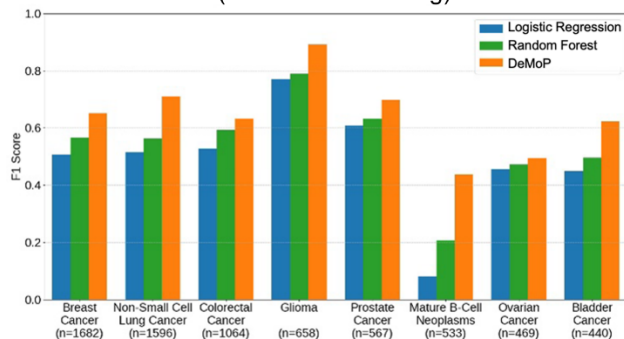
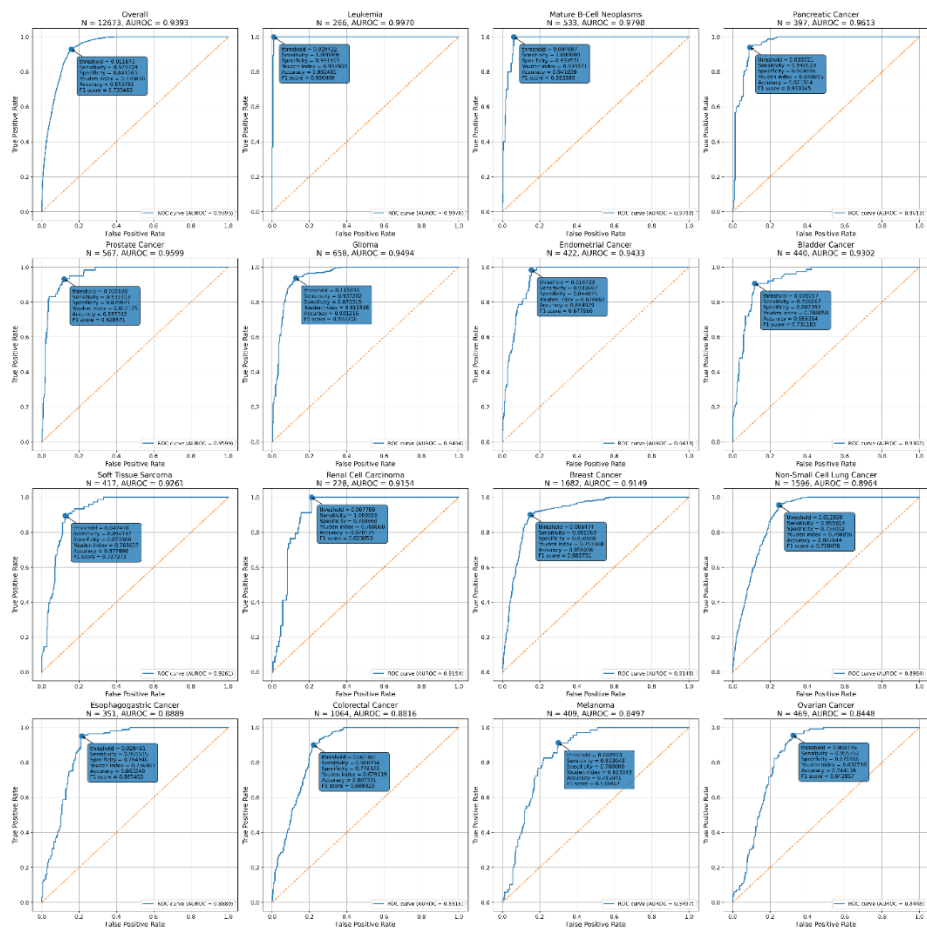
A) Comparison of F1 Score (Class 1) on GENIE dataset**C) Top 20 Genes by Δ Importance (Ablation)****B) F1 Scores by Three Methods for Top 8 Cancer Types (n sorted descending)****D)**

Figure 2. DeMoP performance on the AACR Project GENIE cohort. (A) Class-1 F1 scores on the held-out GENIE test set for DeMoP and all benchmark methods. (B) F1 score improvements across eight cancer types, with DeMoP outperforming random forest in common solid tumors and more than doubling logistic regression's F1 in mature B-cell neoplasms. (C) Top 20 genes ranked by average drop in prediction confidence (Δ) from ablation analysis, highlighting MYC, CDKN2B, ALOX12B and SYK as the most influential candidate genes. (D) Overall and cancer-specific AUROC performance of DeMoP on the GENIE test set.

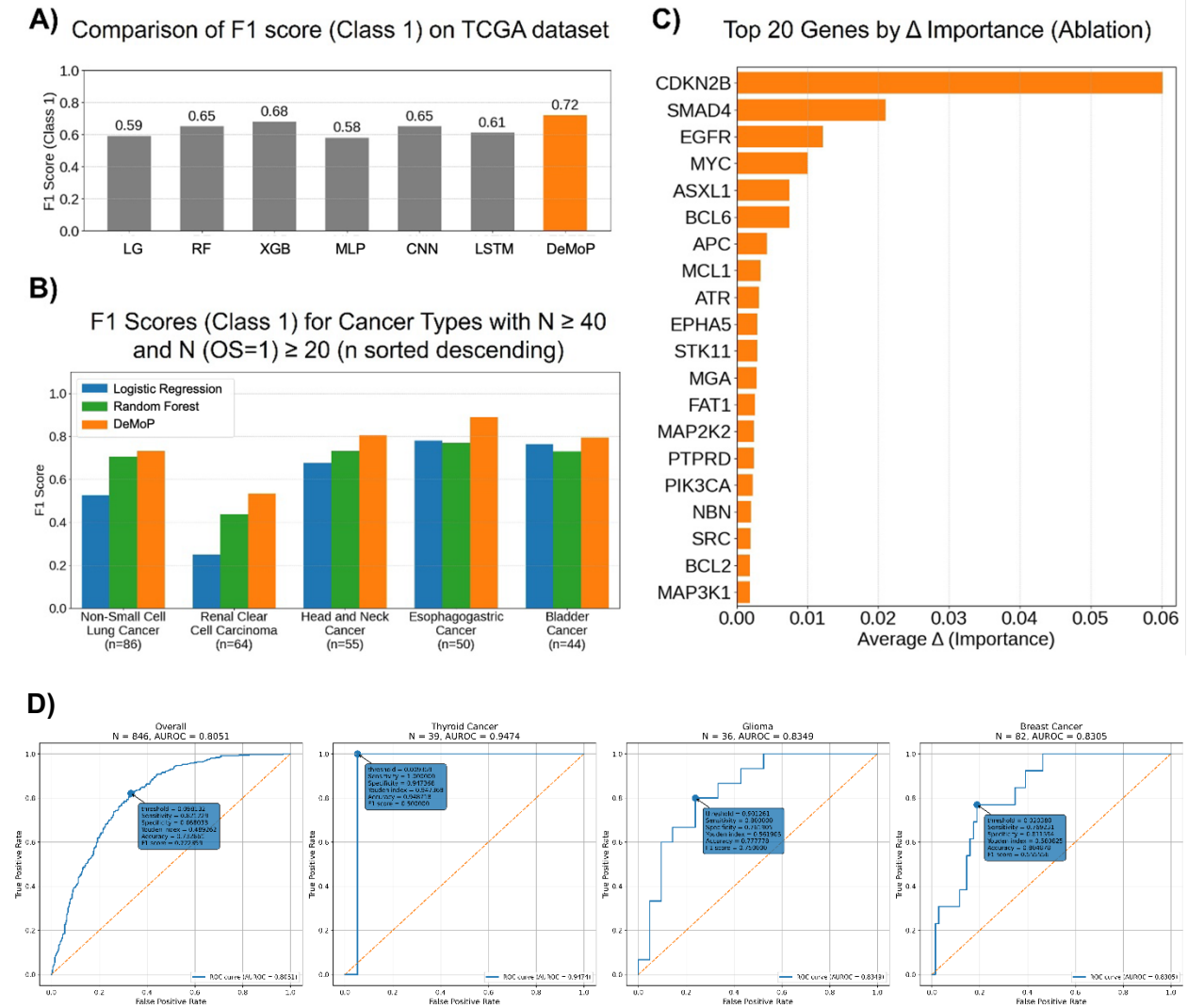


Figure 3. DeMoP performance on the TCGA cohort. (A) Class-1 F1 scores on the held-out test set comparing DeMoP with logistic regression, random forest, XGBoost, CNN, LSTM and MLP. (B) F1 score comparison of DeMoP, random forest and logistic regression across tumour types with at least 40 patients and 20 events (sorted by cohort size). (C) Top 20 genes ranked by average drop in prediction confidence (Δ) from ablation analysis, with CDKN2B, SMAD4, EGFR and MYC among the leading candidate genes. (D) Overall and cancer-specific AUROC performance of DeMoP on the TCGA test set.

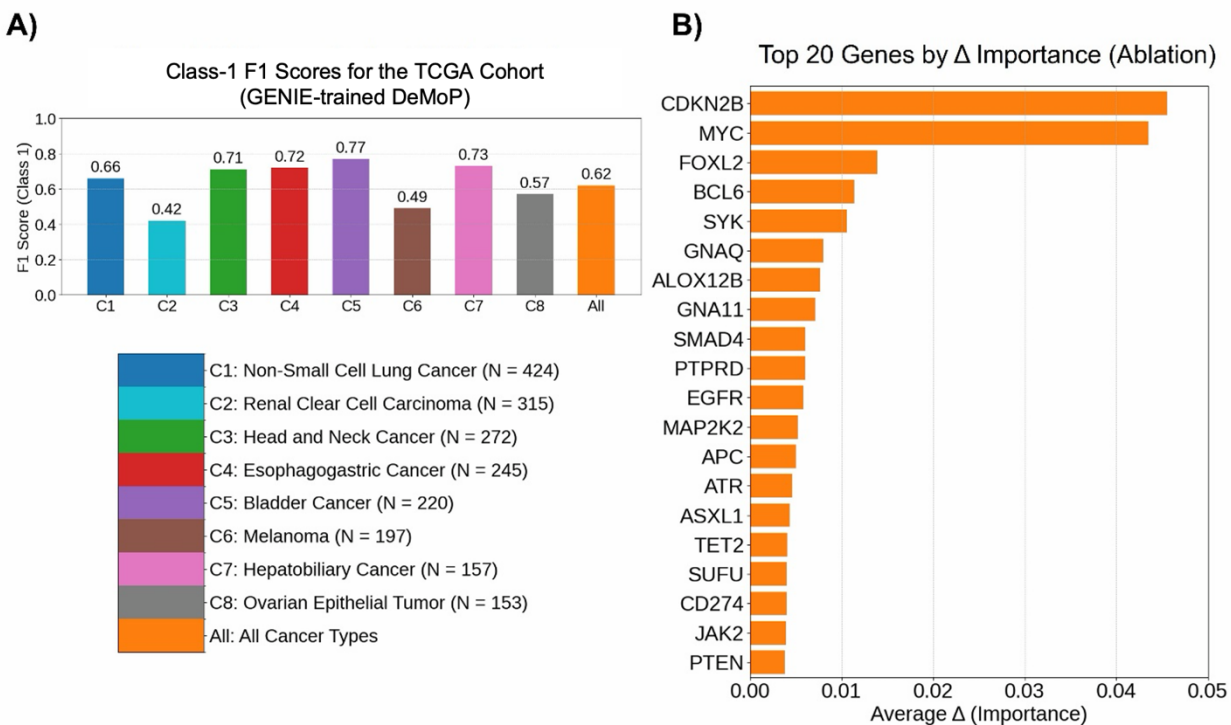


Figure 4. Cross-cohort evaluation of GENIE-trained DeMoP applied to the TCGA cohort. (A) Class-1 F1 scores for each of eight tumour subcohorts (C1–C8), with highest transfer performance in bladder cancer (0.77), hepatobiliary cancer (0.73) and esophagogastric cancer (0.72), moderate transfer in non-small cell lung cancer (0.66) and head and neck cancer (0.71), and lowest transfer in renal clear cell carcinoma (0.42) and melanoma (0.49); overall class-1 F1 = 0.62. (B) Top 20 genes ranked by average drop in prediction confidence (Δ) under cross-cohort ablation, with CDKN2B ($\Delta = 0.05$), MYC ($\Delta = 0.04$), FOXL2 ($\Delta = 0.01$) and SYK ($\Delta = 0.01$) among the leading candidate genes.